

Marcus Gawronsky

APPLIED AI & DECISION SCIENCE · QUANTITATIVE RESEARCH · RESEARCH SOFTWARE

Cape Town, South Africa

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Applied AI and decision-science leader with experience translating mathematical and computational research into production systems, reproducible evidence and reliable software. Built and led technical teams, scaled multimodal inference, designed large-scale causal measurement, and developed a Finance PhD research programme spanning information geometry, representation learning and quantitative finance.

Professional Experience

Solenya

Cape Town, South Africa

CO-FOUNDER & CHIEF TECHNOLOGY OFFICER

Jun. 2024 - Present

- Lead the technical organisation across research, product and infrastructure, including recruitment and team development.
- Built and operated a heterogeneous Kubernetes platform on GCP using Nix, Terraform, Helm and Skaffold, supporting multimodal inference above 250 peak requests per second.
- Designed a randomised experiment spanning nearly one million sessions and validated the commercial effect with complementary causal time-series methods.
- Replatformed product and research surfaces to a Next.js, TypeScript and Storybook monorepo with perfect audited accessibility, best-practice and SEO scores.

MyRunway

Cape Town, South Africa

SENIOR DATA SCIENTIST; DATA SCIENTIST

Jan. 2021 - Apr. 2024

- Built a BigQuery warehouse, CDC and clickstream pipelines, and reusable marts used across pricing, purchasing and stock management.
- Established experimentation and causal-measurement practices and delivered commercially material improvements.
- Fine-tuned and deployed vision-language models and developed optimisation and multi-agent simulation systems for warehouse operations.

DataProphet

Cape Town, South Africa

JUNIOR DATA SCIENTIST

Aug. 2019 - Jan. 2021

- Delivered industrial machine-learning engagements from modelling through deployment and stakeholder handover, while contributing reusable libraries and technical sales material.

WorldQuant University, via Hubble Studios

Cape Town, South Africa

LECTURER & SUBJECT MATTER EXPERT IN FINANCIAL ENGINEERING

Sep. 2018 - Feb. 2019

- Developed graduate financial-engineering material covering risk, simulation, machine learning and macroeconomic modelling.

Technical Competencies

Languages	Python, SQL, TypeScript/JavaScript, R
Econometrics & causal inference	Spatial, network and financial econometrics; experimental design; randomised trials; time-series and difference-in-differences methods
Mathematics & statistics	Functional data analysis; information geometry; optimal transport; graph theory; stochastic optimisation; mixed-integer programming
Machine learning	Deep, representation, multimodal and contrastive learning; language models; explainability; multi-agent simulation
Data & infrastructure	GCP, Kubernetes, Terraform, Nix, Helm; BigQuery, PostgreSQL, ClickHouse, Kafka, Milvus, Redis; dbt, Meltano

Awards

2019	Winner Uber Movement/SANRAL Cape Town Data Science Challenge	Cape Town
2018	Third place Capitec BBLB Kaggle Data Science Competition	South Africa

Research Publications

Submitted / under review	<i>Systematic Covariance Envelopes from Wasserstein Geometry: Evidence from Language-Model Representations</i>
Submitted / under review	<i>Wasserstein-Barycentric Interaction Fields for Spatial Factor Models: Evidence from Language-Model Representations</i>
Submitted / under review	<i>Portfolio Risk Bounds without Cross-Asset Return Covariances: Distributional Fields from Language-Model Representations</i>

Open Source & Technical Writing

Open source	PricingPerspective; precarious-papers; contribution to Hugging Face PEFT
Technical writing	Models are Markup, Tokens are Features; Beyond Average Revenue; Architecting Solenya API V1

Conference Presentations

Jul. 2025	Presented <i>Continuous Risk Factor Models: Analyzing Asset Correlations through Energy Distance</i> , World Finance Conference, Malta
Dec. 2026	Accepted: <i>Hotelling Information Competition in Wasserstein Space: Evidence from Language-Model Representations</i> , World Finance & Banking Symposium, Sharjah

Education

University of Cape Town

Cape Town, South Africa

PHD CANDIDATE IN FINANCE

2022 - Expected 2027

- Research in spatial econometrics, representation learning, functional data analysis and information geometry, focused on the micro-foundations of spatial arbitrage and interpretable financial models.

University of Cape Town

Cape Town, South Africa

MSC ADVANCED ANALYTICS & DECISION SCIENCES

2019 - 2021

- Statistics with Data Science; thesis: *Pricing Offshore Services: Evidence from the Paradise Papers*.

University of Cape Town

Cape Town, South Africa

BBUSc FINANCE, MINOR IN ECONOMICS

2014 - 2018

- Finance, Investments & Banking major; awarded the Class Medal for Natural Resource Economics.

Leadership & Governance

University Governance

University of Cape Town

TREASURER-GENERAL, SRC; SPEAKER, STUDENT PARLIAMENT; TREASURER, COMMERCE STUDENTS' COUNCIL

2015 - 2017

- Served on University finance and student-funding committees and authored the unanimously adopted 2016 UCT Fee Proposal.
- Raised more than R2.5 million through corporate donations and an alumni phonathon for an open student-application programme.

World Universities Debating Championship

University of Cape Town

Co-CONVENER, CAPE TOWN WUDC 2019 BID & DELIVERY

2014 - 2015; 2019

- Helped secure the bid and supported delivery of a 950-participant programme, coordinating staff, volunteers, contractors and funding proposals.